THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In hypermetropia, the near point is 1

(a) at 25 cm **(b)** farther than 25 cm
(c) at infinity **(d)** nearer than 25 cm

2. When white light passes through a prism, the colours from the least bent to the most bent are 1

(a) blue, green, red, yellow, orange, indigo, violet
(b) red, yellow, orange, green, blue, indigo, violet
(c) violet, indigo, blue, green, yellow, orange, red
(d) red, orange, yellow, green, blue, indigo, violet

3. The image formed on the retina is 1

(a) real and inverted **(b)** real and erect
(c) virtual and erect **(d)** virtual and inverted

4. Presbyopia occurs due to 1

(a) weakening of ciliary muscles and loss of flexibility of the lens
(b) long eyeball
(c) short eyeball
(d) damaged cornea

5. A rainbow is formed due to 1

(a) only reflection
(b) only scattering
(c) diffraction
(d) dispersion, refraction and internal reflection by water droplets

6. The least distance of distinct vision for a normal eye is about 1

(a) 2.5 cm **(b)** 25 cm **(c)** 1 m **(d)** 25 m

7. Presbyopia is corrected by 1

(a) plane glass **(b)** prism **(c)** concave mirror **(d)** bifocal lenses

8. A rainbow is seen when the Sun is 1

(a) overhead only **(b)** below the horizon
(c) behind the observer **(d)** in front of the observer

9. The far point of a normal human eye is 1

(a) 100 m **(b)** at infinity **(c)** 1 m **(d)** 25 cm

10. A clouding of the eye lens that causes blurred vision in old age is 1

(a) night blindness **(b)** cataract **(c)** astigmatism **(d)** myopia

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the defect of vision in which the person cannot see nearby objects clearly. Give a cause. 2

12. Why do stars twinkle? 2

13. What is the far point and near point of a normal eye? 2

14. What is presbyopia? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What is myopia? State two causes and the corrective lens. 3

16. What is dispersion? Name the colours in the order of the spectrum. 3

17. Why is the colour of the clear sky blue? 3

OR

What is hypermetropia? State two causes and the corrective lens. 3

18. Explain why white light splits into colours in a prism. 3

OR

Why does the sky appear dark to an astronaut? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain myopia: cause, effect and correction, with the help of a ray diagram description. 5

OR

Explain atmospheric refraction with examples: twinkling of stars and advance sunrise. 5

20. Explain hypermetropia: cause, effect and correction with the help of a ray diagram description. 5

OR

Explain scattering of light and why the sky is blue and the sun appears red at sunrise and sunset. 5

END OF QUESTION PAPER — Total: 40 Marks